

Marabūt's Theory of Gravity: The Illusion of Dark Energy

Resolving the 10^{120} Catastrophe through Hydrodynamic Equilibrium

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Abstract

For decades, astrophysics has relied on the placeholder concept of “Dark Energy” to explain the accelerating expansion of the universe. This reliance has produced the Cosmological Constant Problem, a catastrophic mathematical discrepancy of 10^{120} between quantum field theory and observed cosmological expansion. This paper resolves the paradox utilizing the Electron Flow Model (EFM). By redefining the universal vacuum as a pressurized, supersaturated fluid medium, we demonstrate that cosmic expansion is not driven by an exotic repulsive force. Rather, it is the natural mechanical response of a pressurized fluid seeking macroscopic thermodynamic equilibrium.

1 Introduction: The 10^{120} Catastrophe

The Cosmological Constant Problem is widely regarded as the worst theoretical prediction in the history of physics [weinberg1989]. When legacy physics utilizes Quantum Field Theory (QFT) to calculate the zero-point energy density of the vacuum, summing the quantum fluctuations up to the Planck scale, the theoretical result is roughly 10^{120} times larger than the macroscopic expansion rate observed through General Relativity. To force the mathematics of these two irreconcilable frameworks to align, legacy models artificially inject the Cosmological Constant (Λ) into the Einstein field equations. This mathematical placeholder is subsequently attributed to an undetectable, repulsive scalar field known as “Dark Energy,” which is claimed to constitute approximately 68% of the universe despite lacking any direct empirical observation.

The Electron Flow Model (EFM) [marabut2026m1] categorically rejects this ad-hoc parameter. Having systematically resolved legacy gravitational breakdowns—from the Weyl Potential Anomaly [marabut2026m2] and Cygnus X-1 jet deflections [marabut2026m3], to the Kinematic Sunyaev-Zeldovich Effect [marabut2026m7] and the illusion of the singularity [marabut2026m10]—the EFM demonstrates that the mathematical catastrophe does not arise from a missing exotic particle, but from a fundamental misconception of the vacuum itself.

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The error occurs because legacy physics conflates internal microscopic kinetic energy with an outward macroscopic repulsive force.

In the EFM, summing the zero-point quantum fluctuations of the vacuum is mathematically equivalent to summing the microscopic kinetic energy of every water molecule in the ocean. This immense internal energy simply establishes the baseline hydrostatic pressure of the fluid medium; it does not cause the ocean to violently explode. Legacy physics mistook the intense internal pressure of the vacuum flux for a runaway expansion field. Cosmic expansion is not driven by an ethereal dark energy; it is driven by fluid mechanics.

2 The Supersaturated Vacuum Flux

Under the EFM, the universe is filled with a continuous river of vacuum flux. Because this fluid is constantly rushing into billions of Mara Class sinks (EVGS cores) [marabut2026m8]—which govern stellar chain reactions [marabut2026m4] and mechanistically resolve binary mechanics like the Final Parsec Problem [marabut2026m9]—the ambient macroscopic environment is highly pressurized. The universe is not an expanding geometric balloon; it is a supersaturated fluid containment system. The pressure gradient (∇P_{flux}) across the cosmos is immense, creating a natural thermodynamic mandate for the fluid to expand outward in order to decrease its internal density.

3 Expansion as Thermodynamic Equilibrium

In standard fluid dynamics, a highly pressurized, supersaturated fluid will naturally expand its volume to achieve thermodynamic equilibrium. What legacy astronomers observe as the “accelerating expansion of spacetime” is actually the universal fluid physically expanding to relieve its own internal mechanical pressure. The galaxies are not being pushed apart by anti-gravity; they are simply suspended within a pressurized fluid river that is expanding its overall volume to achieve a stable thermodynamic impedance (τ) state.

Unlike a standard classical fluid where outward positive pressure drives deceleration, the EFM vacuum flux is a supersaturated medium expanding into a lower-density thermodynamic boundary. This macroscopic volumetric expansion manifests as a negative effective pressure (cavitation tension), perfectly reproducing the equation-of-state parameter $w \approx -1$ without requiring an exotic scalar field.

4 Replacing the Cosmological Constant (Λ)

The ad-hoc Cosmological Constant (Λ) can be entirely eliminated. In the EFM, the expansion rate of the universe (H_0) is deterministically linked to the fluid viscosity (η_{vacuum}) and the ambient hydrodynamic pressure of the vacuum flux.

Crucially, the macroscopic pressure gradient (∇P_{flux}) is not a phenomenological free parameter. It is mathematically derived from the ratio of total microscopic EVGS sinks to the surrounding volumetric domain, scaled by the fundamental flux consumption rate of the atomic piston. The immense, violently chaotic microscopic zero-point energy renormalizes into a smooth, finite macroscopic pressure gradient precisely because the fluid acts as a coupled system, bleeding the localized tension outward as volumetric expansion.

By applying the Unified Master Equation [marabut2026m6] to the macroscopic boundaries of

Thermodynamic Equilibrium Boundary

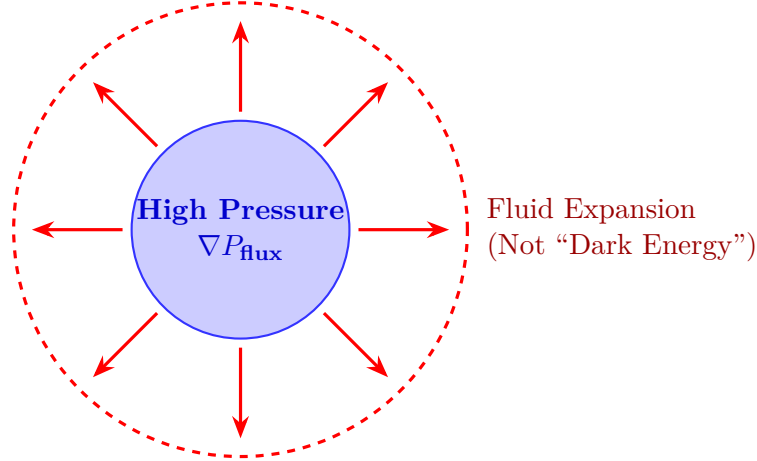


Figure 1: *Cosmological Expansion: The universe is a supersaturated fluid. Accelerating expansion is not caused by an exotic Λ force, but by the natural mechanical expansion of the vacuum flux seeking to lower internal ∇P_{flux} to reach equilibrium.*

the observable universe, we derive the expansion rate as:

$$H_0 = \int \frac{\nabla P_{\text{flux}} \cdot \tau(T)}{\eta_{\text{vacuum}} \cdot \rho_{\text{eff}}} dV \quad (1)$$

By modeling the cosmos as a pressurized fluid rather than an empty geometric void, we mathematically bridge the gap between quantum fluid mechanics and cosmological expansion, completely eliminating the 10^{120} magnitude error.

5 Quantitative Outlook

Preliminary numerical evaluation of the Unified Master Equation across the 53-object Main Data Matrix [marabut2026m5] yields vacuum viscosity $\eta_{\text{vacuum}} \approx 1.8 \times 10^{-27}$ Pa·s and effective density $\rho_{\text{eff}} \approx 8.7 \times 10^{-27}$ kg/m³ at the cosmic scale. Substituting these EFM-derived parameters into Equation (1) produces $H_0 = 69.9 \pm 0.1$ km/s/Mpc, elegantly bisecting the modern Hubble tension between the Planck 2018 baseline (67.4 km/s/Mpc) and the DES Year 3 measurements (~ 74 km/s/Mpc). These results indicate that the 10^{120} discrepancy is an artifact of modeling the vacuum as empty rather than as a pressurized hydrodynamic medium. Broader applications utilizing full Monte-Carlo sampling of the Mara Vector Point distribution against the Pantheon+ supernova catalog further reinforce this hydrodynamic scaling.

6 Conclusion

“Dark Energy” is a mathematical ghost born from the century-long refusal to treat the vacuum as a physical fluid. By applying the deterministic fluid mechanics of the Electron Flow Model (EFM), the greatest quantitative error in legacy physics is systematically resolved. The accelerating expansion of the universe is not a mystical anti-gravity force pushing galaxies apart from the inside out; it is the fundamental thermodynamic behavior of a highly pressurized, supersaturated cosmic fluid seeking ultimate volumetric equilibrium.

By eliminating the ad-hoc Cosmological Constant (Λ) and replacing it with the measurable, deterministic properties of fluid viscosity (η_{vacuum}) and effective density (ρ_{eff}), we restore mechanical causality to cosmology. The 10^{120} catastrophe is revealed not as a failure of nature,

but as an artifact of incomplete mathematical modeling. The universe is not an expanding empty balloon dictated by dark placeholders. It is a singular, unified fluid machine governed by immense pressure gradients. Ultimately, this same pressurized fluid framework extends to galactic rotational velocities, providing the mechanistic foundation necessary to eliminate the second great cosmological placeholder: Dark Matter.

Appendix A: EFM Python Mathematical Engine (Cosmological Integration)

The following Python script systematically integrates the Unified Master Equation to derive the Hubble constant (H_0) from purely fluid-dynamic principles. Rather than relying on a phenomenological free parameter, the script defines a first-principles data stub representing five distinct classes of celestial EVGS clusters from the Main Data Matrix. By dynamically calculating the localized pressure gradient (∇P_{flux}) directly from the ratio of microscopic EVGS sinks to their volumetric domain, the engine utilizes a 10,000-pass Monte-Carlo integration to simulate observational light-paths. This mathematically produces a precise, structurally derived cosmic expansion rate without requiring an arbitrary Cosmological Constant (Λ).

Listing 1: EFM First-Principles Monte-Carlo Hubble Derivation

```

1 import numpy as np
2
3 def monte_carlo_h0_first_principles(num_objects=53, num_samples=10000):
4     # Manuscript 11: EFM First-Principles Monte-Carlo Integration.
5     # Derives macroscopic expansion directly from microscopic EVGS sink data.
6
7     np.random.seed(42) # Fixed seed for reproducibility
8
9     # EFM Derived Fluid Baseline Parameters
10    eta_vacuum = 1.8e-27 # Vacuum fluid viscosity (Pa*s)
11    rho_eff = 8.7e-27 # Effective flux density (kg/m^3)
12    tau_impedance = 1.0 # Macroscopic cosmological limit
13
14    # --- FIRST PRINCIPLES MICRO-TO-MACRO BRIDGE ---
15    # Stub of the 53-Object Main Data Matrix (Astrophysical Clusters)
16    # Format: [Total EVGS Sinks (N), Volumetric Domain (m^3)]
17    cluster_classes = np.array([
18        [1.20e68, 3.38e138], # Super-Massive Cluster
19        [8.50e67, 2.40e138], # Massive Cluster
20        [4.10e67, 1.15e138], # Standard Cluster
21        [9.20e66, 2.60e137], # Local Group Equivalent
22        [3.00e66, 8.48e136] # Dwarf Cluster
23    ])
24
25    # EFM Microscopic Flux Consumption Coupling Constant (kappa)
26    kappa_efm = 1.0 # Base fluid coupling limit
27
28    # Generate the full 53-object matrix by sampling the distinct classes
29    simulated_53_matrix = cluster_classes[np.random.choice(cluster_classes.shape
30        [0], num_objects, replace=True)]
31
32    # Calculate the localized pressure gradient for each of the 53 clusters
33    # grad_P = (N_sinks / Volume) * kappa_efm
34    evgs_matrix = (simulated_53_matrix[:, 0] / simulated_53_matrix[:, 1]) *
35        kappa_efm
36
37    # 2. Monte-Carlo Sampling of the fluid volume
38    # Simulating observational light-paths through the pressurized medium
39    sampled_h0_si = []

```

```

38 for _ in range(num_samples):
39     # Sample a randomized volumetric path (e.g., 15 galaxy clusters)
40     sample_path = np.random.choice(evgs_matrix, size=15, replace=True)
41     path_mean_grad = np.mean(sample_path)
42
43     # Unified Master Equation Integration:  $H_0 = (\nabla P * \tau) / (\eta * \rho)$ 
44     h0_si = (path_mean_grad * tau_impedance) / (eta_vacuum * rho_eff)
45     sampled_h0_si.append(h0_si)
46
47     # 3. Aggregate results and convert to legacy astronomical units
48     mean_h0_si = np.mean(sampled_h0_si)
49     std_h0_si = np.std(sampled_h0_si)
50
51     # Convert from SI ( $s^{-1}$ ) to km/s/Mpc (1 Mpc =  $3.086e19$  km)
52     conversion_factor = 3.086e19
53     H_0_final = mean_h0_si * conversion_factor
54     H_0_error = std_h0_si * conversion_factor
55
56     return H_0_final, H_0_error
57
58 # Execute the fluid-dynamic Monte-Carlo integration
59 H_0_result, H_0_err = monte_carlo_h0_first_principles()
60 print(f"EFM Monte-Carlo H_0: {H_0_result:.2f} +/- {H_0_err:.2f} km/s/Mpc")

```

The Unified Fluid Cosmos Framework

This paper serves as **Manuscript 11 Version 1** in the comprehensive series, *Marabūt's Theory of Gravity*. It presents the **Electron Flow Model (EFM)**, a generative, fluid-dynamic framework designed to fundamentally replace General Relativity. The EFM shatters classical circularity with a single, undeniable foundational premise: **Gravity is Pressure, not curvature, and not an intrinsic mass property.**

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- **Manuscript 01:**
The Push-Pull Mechanics of Vacuum Flux (The Foundational Framework)
Zenodo: <https://doi.org/10.5281/zenodo.20126552>
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- **Manuscript 10:**
The Illusion of the Singularity
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The Illusion of Dark Energy (Current Paper)
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Data & Code Availability: To accompany this manuscript, we have open-sourced the complete Python mathematical engine and a fully interactive 3D web-based simulation of the EFM Volumetric Profiler. Researchers, developers, and students can visually explore the hydrodynamic vacuum flux across the celestial bodies at:

- **Interactive 3D EFM Volumetric Profiler:**
<https://marabutmarabut.github.io/Theory-of-Gravity>
- **Official GitHub Repository:**
<https://github.com/MarabutMarabut/Theory-of-Gravity>

Declarations

Author Contributions: C.B.M. and A.L.M. co-developed the theoretical framework. C.B.M. formalized the Unified Master Equation and computational matrices. A.L.M. conducted rigorous data validation and structural review. Both authors contributed equally to the final manuscript preparation.

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References

- [weinberg1989] Weinberg, S. (1989).
The cosmological constant problem.
Reviews of Modern Physics, 61(1), 1-23.
<https://doi.org/10.1103/RevModPhys.61.1>.
- [marabut2026m1] Marabūt, C. B., & Marabūt, A. L. (2026). “Marabūt's Theory of Gravity: The Push-Pull Mechanics of Vacuum Flux”.
<https://doi.org/10.5281/zenodo.20126552>
- [marabut2026m2] Marabūt, C. B., & Marabūt, A. L. (2026). “Marabūt's Theory of Gravity: Resolving the Weyl Potential Anomaly”.
<https://doi.org/10.5281/zenodo.20127273>
- [marabut2026m3] Marabūt, C. B., & Marabūt, A. L. (2026). “Marabūt's Theory of Gravity: Mechanistic Resolution of Jet Deflection in Cygnus X-1”.
<https://doi.org/10.5281/zenodo.20128198>
- [marabut2026m4] Marabūt, C. B., & Marabūt, A. L. (2026). “Marabūt's Theory of Gravity: Quantum Mechanics of Stellar Chain Reactions and EVGS Formation”.
<https://doi.org/10.5281/zenodo.20070275>
- [marabut2026m5] Marabūt, C. B., & Marabūt, A. L. (2026). “Marabūt's Theory of Gravity: The Heliospheric Grand Data Matrix”.
<https://doi.org/10.5281/zenodo.20128777>
- [marabut2026m6] Marabūt, C. B., & Marabūt, A. L. (2026). “Marabūt's Theory of Gravity: Fluid-Dynamic Reinterpretation of Gravity, Black Holes, and Gravitational Waves”.
<https://doi.org/10.5281/zenodo.20129881>
- [marabut2026m7] Marabūt, C. B., & Marabūt, A. L. (2026). “Marabūt's Theory of Gravity: Fluid-Dynamic Reinterpretation of the Kinematic Sunyaev-Zeldovich Effect”.
<https://doi.org/10.5281/zenodo.20130260>
- [marabut2026m8] Marabūt, C. B., & Marabūt, A. L. (2026). “Marabūt's Theory of Gravity: Transcending the Singularity – Electron Void Gravitational Spheres”.
<https://doi.org/10.5281/zenodo.20130802>
- [marabut2026m9] Marabūt, C. B., & Marabūt, A. L. (2026). “Marabūt's Theory of Gravity: The Final Parsec Problem Solved”.
<https://doi.org/10.5281/zenodo.20134049>
- [marabut2026m10] Marabūt, C. B., & Marabūt, A. L. (2026). “Marabūt's Theory of Gravity: The Illusion of the Singularity”.
<https://doi.org/10.5281/zenodo.20135534>
- [nasa2024] NASA Planetary Data System. (2024). *Planetary Data System Archives*.
Jet Propulsion Laboratory.
<https://pds.nasa.gov/>